

Due to an issue with AWS billing card verification, the task was completed in the LocalStack AWS-compatible environment using the AWS CLI, S3 emulation, and local equivalents of Glue/Athena.

```
PS D:\AWS_Lab> aws --endpoint-url=http://localhost:4566 s3 ls
2026-05-12 20:54:19 test-bucket
PS D:\AWS_Lab> aws --endpoint-url=http://localhost:4566 s3 mb s3://data-lake-bucket
make_bucket: data-lake-bucket
PS D:\AWS_Lab> aws --endpoint-url=http://localhost:4566 s3 ls
2026-05-12 20:54:19 test-bucket
2026-05-12 21:09:49 data-lake-bucket
PS D:\AWS_Lab> aws --endpoint-url=http://localhost:4566 s3 cp data.csv s3://data-lake-bucket
upload: .\data.csv to s3://data-lake-bucket/data.csv
PS D:\AWS_Lab> aws --endpoint-url=http://localhost:4566 s3 ls s3://data-lake-bucket
2026-05-12 21:10:31      1130 data.csv
PS D:\AWS_Lab> |
```

Figure 1 – LocalStack S3

```
PS D:\AWS_Lab> aws --endpoint-url=http://localhost:4566 iam create-user --user-name aws_admin
{
  "User": {
    "Path": "/",
    "UserName": "aws_admin",
    "UserId": "feefd5tnf7ut7t8zj87d",
    "Arn": "arn:aws:iam::000000000000:user/aws_admin",
    "CreateDate": "2026-05-12T18:11:54.488000+00:00"
  }
}
PS D:\AWS_Lab> aws --endpoint-url=http://localhost:4566 iam list-users
{
  "Users": [
    {
      "Path": "/",
      "UserName": "aws_admin",
      "UserId": "feefd5tnf7ut7t8zj87d",
      "Arn": "arn:aws:iam::000000000000:user/aws_admin",
      "CreateDate": "2026-05-12T18:11:54.488000+00:00"
    }
  ]
}
PS D:\AWS_Lab> |
```

Figure 2 – CLI IAM

```
PS D:\AWS_Lab> python schema.py

=== TABLE STRUCTURE ===
employee_id      int64
name              object
department        object
position          object
salary           int64
experience_years  int64
city              object
dtype: object

=== SAMPLE DATA ===
   employee_id  name department position  salary  experience_years  city
0            1  Anna Kovalenko      IT   Data Engineer    2200              3   Kyiv
1            2   Ivan Melnyk       HR    HR Manager    1400              5   Lviv
2            3  Oleh Shevchenko  Finance  Financial Analyst    1800              4   Kyiv
3            4   Maria Bondar      IT   Backend Developer    2500              6  Kharkiv
4            5  Petro Tkachenko  Marketing  SEO Specialist    1300              2   Odesa
PS D:\AWS_Lab> |
```

Figure 3 – Python schema inference

```

PS D:\AWS_Lab> python athena_query.py
employee_id      name department      position      salary  experience_years  city
0               1   Anna Kovalenko    IT      Data Engineer    2200              3   Kyiv
1               2     Ivan Melnyk      HR      HR Manager       1400              5   Lviv
2               3   Oleh Shevchenko   Finance Financial Analyst   1800              4   Kyiv
3               4     Maria Bondar     IT      Backend Developer  2500              6   Kharkiv
4               5   Petro Tkachenko   Marketing SEO Specialist    1300              2   Odesa
5               6     Sofiia Koval      IT      Data Analyst      2000              3   Dnipro
6               7     Andrii Mazur       Sales   Sales Manager     1700              5   Ternopil
7               8   Nataliia Hrytsenko Finance Accountant        1600              4   Lviv
8               9     Dmytro Savchenko   IT      DevOps Engineer   2800              7   Kyiv
9              10     Olena Kravets      Marketing Content Manager    1250              2   Odesa
PS D:\AWS_Lab> |

```

Figure 4 – DuckDB SQL